

Appendix 3.3: Specification of Semantic AI Oracles (TRANSCEPTOR AI ORACLES)

1. Purpose and System Scale of Agora Semantic AI-Oracle

- **Module Identifier:** `MOD-AGO-CME-v3.4-Adaptive`.
- **Execution Stack:** C++17 / Python FastAPI / ONNX Runtime / Open Enclave SDK inside TEE Enclave.
- **PoUCC Standard (Proof of Useful Cognitive Computation):** Algorithmic standard for network compute allocation. Under this standard, **75% of network GPU resources** are dedicated to high-performance linguistic and vector calculations, NLP analysis of normative acts, commercial estimate auditing for the B2B sector, and de-anonymization of cartel fraud. The remaining 25% are divided between DID infrastructure maintenance (20%) and MPC connectivity protocol support (5%).

Component	Tech Stack	Function in CME Contour
Compute Kernel	C++17 / Open Enclave SDK	Execution of critical algorithms inside TEE enclave
API & Service Layer	Python FastAPI / ONNX Runtime	Asynchronous vector parsing and local model execution
Compute Allocation	PoUCC (75% GPU Clusters)	Semantic decompilation, vector estimate audit, anti-fraud

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2. Three-Layer Deconstruction and Admission Threshold Filter

Prior to being submitted for public voting, every civil or institutional initiative undergoes mandatory automated pre-moderation through a decentralized network of Semantic AI Oracles. Pre-moderation performs a three-layer vector decompilation of project text and cost estimate documentation:

- **Layer 1: Jurisprudence ($Comp_{law}$):** Evaluation of proposal text for compliance with the Fundamental Human Rights Contour, existing legislative framework, and Whitepaper

system invariants. Automated detection of hidden legal trapdoors, jurisprudential contradictions, and corruption loopholes is executed.

- **Layer 2: Budget Estimates ($Comp_{fin}$):** Vector parsing of all financial metrics, estimate specifications, and budgetary allocation requirements using cosine similarity methods. This layer flags populist cost inflation, unbudgeted expenditure requests, and signs of cartel price-fixing.
- **Layer 3: Ecology ($Comp_{eco}$):** Direct mathematical modeling of the project's comprehensive ecological footprint. Carbon, acoustic, hydrological, and thermodynamic impact parameters on the living environment are calculated.

Threshold Admission Filter (Heaviside Step Filter):

The final admission threshold is calculated as the product of Heaviside step functions across all three layers:

$$\Psi = Step(Comp_{law} - Th_{law}) \cdot Step(Comp_{fin} - Th_{fin}) \cdot Step(Comp_{eco} - Th_{eco})$$

where Th_{law}, Th_{fin}, and Th_{eco} are strictly established threshold values of the corresponding compliance standards.

- If $\Psi = 1.0$, the initiative is recognized as valid and transferred to the general voting contour.
- If $\Psi = 0$, the initiative is processed according to the Cognitive Mediation Protocol.

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3. Cognitive AI-Architect (AI-Architect Engine)

The module eliminates the technocratic barrier and citizen alienation from governance participation, transforming the Semantic AI Oracle from a censor into a project design institute. Upon receiving a civil initiative lacking a formalized feasibility study or cost estimates ($\Psi = 0$), the system initiates the auto-completion procedure:

1. **Extraction of Value Core:** Cleansing the text of emotional, manipulative, and populist syntax.
2. **Generation of 3-Layer ScenarioBrief:**
    - *Layer 1 (Jurisprudence):* Alignment with Whitepaper invariants and fundamental human rights.
    - *Layer 2 (Estimates):* Automatic budget calculation in T_{net} tokens based on benchmark RWA databases and market indicators.
    - *Layer 3 (Ecology):* Thermodynamic and carbon footprint modeling.
3. **Co-Authorship & One-Click Approval:** Delivery of the compiled package back to the author for one-click approval via the Personal Companion (Satellite Bridge), after which the project enters the Agora with a compliance rating of $\Psi = 1.0$.

4. Contour Boundary Enforcer & Atomic Proposal Split

Upon receiving proposals, the `Contour Boundary Enforcer` module enforces a strict separation of decision-making logic and voting rights:

- **Value Contour (Ethics, Rights, Culture):** Applies the Cleisthenic Absolute Isonomy regime. The effective voting weight of any resident is strictly fixed at $W_{effective} = 1.0$. Meritocratic surcharges and capital are forcibly zeroed out, and Quadratic Voting (QV) is hardware-blocked at the kernel level.
- **Mathematical / Economic Contour (Resources, Estimates, IT):** Applies Socratic Dynamic Meritocracy. Voting weight is bounded within the interval $W_{user} \in [0.01; 5.00]$ based on Proof-of-Contribution. Capital-pool dominance is suppressed by the Quadratic Voting cost progression $C(V) = V^2$.

Upon an attempt to disguise a budget-intensive economic proposal under the guise of a humanitarian manifesto, the module executes an **Atomic Proposal Split**:

1. Humanitarian Manifesto → Routed to Value Contour ($W_{effective} = 1.0$, No QV).
2. Executive Feasibility Study and Estimate → Routed to Mathematical Contour ($W_{user} \in [0.01; 5.00]$, $C(V) = V^2$, 3-layer `ScenarioBrief`).

Financial release is unlocked strictly upon the approval of both linked acts.

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5. Affiliation Detector and Guild Protection (Human-in-the-Loop)

The Semantic AI Oracle continuously executes vector parsing of estimate documentation using cosine similarity to detect signs of covert cartel collusion and specific "trapdoors" in technical specifications. Upon detecting affiliation, required collateral increases exponentially ($C(V) = V^2$), and contract profit margin is discounted in favor of the Regeneration Reserve Fund ($S_{reserve}$).

Under the Human-in-the-Loop framework, atypical edge cases and appeals are escalated to the arbitration contour of meritocratic Guilds:

- **Double-Blind Rotation:** Selection of expert arbitration panels is executed via randomized rotation based on Zero-Knowledge Commitments (ZK-Commitments). Experts and authors remain fully anonymized from one another.
- **NLP Style Sanitizer:** Reports and evaluation materials are subjected to mandatory stripping of stylistic quirks, jargon, and formatting prior to expert review.

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6. Adaptive Mechanisms v3.4-Adaptive

- **Two-Level Gradient Filter "Physics vs Failure":** Any telemetry anomaly ($|M_{fact} - M_{target}| > Threshold$) shifts the transaction to `PENDING` status. The Edge AI agent initiates cross-triangulation of adjacent IoT sensors to calculate the Environmental Correlation Coefficient (K_{env}). If $K_{env} \geq Threshold_{env}$ (physical anomaly, natural disaster), the Adaptation Buffer is activated without applying sanctions. If $K_{env} < Threshold_{env}$ (deliberate falsification), the "Red Button" protocol is triggered — collapsing vote weight to $W_{user} = 0.01$ and seizing collateral.
- **Proof-of-Useful-Outcome (PoUO) Utility Contour:** The platform is fully shielded against Goodhart's Law by rejecting static formal activity in Scopus, ORCID, or GitHub. Discrete increments ($\Delta I_{cp} = +0.01$) are credited strictly upon verification of real socio-infrastructural or ecological outcomes via IoT telemetry.